

Benjamin Kaminow

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Research Interests

My current research focuses on evaluating structure-based graph neural networks for protein-ligand binding affinity prediction in the context of rational small molecule drug design. Specifically, I am interested in exploring why structure-based models underperform relative to ligand-only models, with an emphasis on physical biases such as E(3)-invariance and -equivariance. More generally, I am excited by any research that combines structural biology, physics, and machine learning.

Education

Ph.D. in Computational Biology and Medicine

July 2021 – Sept 2026 (Scheduled)

Memorial Sloan Kettering Cancer Center
and Weill Cornell Medical College, New York, NY

Advisor: John Chodera

Thesis: “Structure-based graph machine learning methods for protein-ligand binding affinity prediction”

B.S. in Chemical and Biomolecular Engineering

Sept 2015 – May 2019

The Johns Hopkins University, Baltimore, MD

Research Experience

The Chodera Lab, Memorial Sloan Kettering Cancer Center

Jan 2022 – Present

Graduate Student

- Investigating the importance of incorporating physical biases in structure-based machine learning models for protein-ligand binding affinity prediction
- Assessing the performance, data efficiency, and generalizability of structure-based graph neural networks in the task of predicting protein-ligand binding affinities
- Developing a framework for modularly adapting arbitrary neural network architectures for structure-based machine learning
- Supported the drug discovery efforts of the AI-driven Structure-enabled Antiviral Platform through development of a computational chemistry software suite designed to assist at key stages of the drug discovery pipeline
- Conceptualized and planned appropriately scoped projects for rotation graduate students and summer undergraduate interns, and mentored these students during their time in the lab
- Implemented an alternative atom enumeration scheme in the **espaloma** molecular machine learning model

The Dobin Lab, Cold Spring Harbor Laboratory

June 2019 – June 2021

Computational Science Developer

- Developed the ConsDB Python package to evaluate functional effects of replacing the current reference genome with different consensus genomes using the 1000 Genomes Project
- Analyzed scRNA-seq and ChIP-seq data to study differential splicing and epigenetic regulation

The Karchin Lab, Institute for Computational Medicine

June 2018 – May 2019

Undergraduate Research Assistant

- Improved the MHCnuggets machine learning model for peptide-MHC binding affinity prediction by improving allele matching for rare alleles and exploring more biologically relevant features
- Evaluated the ability of the MHCnuggets pipeline to successfully predict cancer patient survival based on MHC peptide presentation

The Betenbaugh Lab,

Jan 2017 – Sept 2018

Advanced Mammalian Biomanufacturing Innovation Center

Undergraduate Research Assistant

- Developed and utilized MATLAB and Python tools for Flux Balance Analysis to perform *in silico* optimization of Chinese hamster ovary cell growth medium

Awards and Honors

- National Science Foundation Graduate Research Fellowship Program, 2023
- Graduated from Johns Hopkins University with General Honors and Departmental Honors, 2019
- Johns Hopkins University Dean's List, Fall 2015-Spring 2016, Spring 2017-Spring 2019

Skills

Machine Learning/Deep Learning: Hyperparameter tuning, model architecture selection, pre-training, data processing, graph neural networks, PyTorch, e3nn, PyTorch Geometric, Weights & Biases, scikit-learn

Computational Chemistry: Structure visualization, molecular docking, OpenEye Toolkits, RDKit, PyMOL

Scientific Computing: High-performance computing, Git, Linux, data visualization, NumPy, Pandas, Matplotlib, Seaborn, SciPy

Programming Languages: Python, Bash, R, Rust, C++

Publications

1. **Kaminow, B.**, Payne, A. M., MacDermott-Opeskin, H., Singh, S., Chodera, J. D., "Physical biases improve performance of structure-based binding affinity models," *Draft available upon request*, 2026.
2. Payne, A. M., **Kaminow, B.**, MacDermott-Opeskin, H., Pulido, I., Scheen, J., Castellanos, M. A., Fearon, D., Chodera, J. D., Singh, S., "How many crystal structures do you need to trust your docking results?," *J Med Chem*, *in press*, 2026.
3. MacDermott-Opeskin, H., Scheen, J., Wognum, C., ... **Kaminow, B.**, ... Chodera, J. D., "A Computational Community Blind Challenge on Pan-Coronavirus Drug Discovery Data," *J Chem Inf Model*, 2026.
4. López-Ríos de Castro, R., Rodríguez-Guerra, J., Schaller, D., ... **Kaminow, B.**, ... Volkamer, A., Chodera J. D., "The Journey of Data: Lessons Learned in Modeling Kinase Affinity, Selectivity, and Resistance [Article v1.0]," *Living J Comput Mol Sci*, 2025.
5. Boby, M. L., Fearon, D., Ferla, M., Mihajlo, F., Koekemoer, L., Robinson, M. C., **The COVID Moonshot Consortium**, Chodera, J. D., Lee, A., London, N., von Delft, A., von Delft, F., "Open science discovery of potent noncovalent SARS-CoV-2 main protease inhibitors," *Science*, 2023.
6. Wang, Y., Fass, J., **Kaminow, B.**, Herr, J. E., Rufa, D., Zhang, I., Pulido, I., Henry, M., Bruce Macdonald, H. E., Takaba, K., Chodera, J. D., "End-to-end differentiable construction of Molecular Mechanics Force Fields," *Chem Sci*, 2022.
7. **Kaminow, B.**, Ballouz, S., Gillis, J., Dobin, A., "Pan-human consensus genome significantly improves the accuracy of RNA-seq analyses," *Genome Res*, 2022.
8. Shao, X. M., Bhattacharya, R., Huang, J., Sivakumar, I. A., Tokheim, C., Zheng, L., Hirsch, D., **Kaminow, B.**, Omdahl, A., Bonsack, M., Riemer, A., Velculescu, V., Anagnostou, V., Pagel, K., Karchin, R., "Prediction of MHC Class I and II Neoantigens with MHCnuggets," *Cancer Immunol Res*, 2019.

Invited Talks

1. **Kaminow, B.**, "Model construction for structure-based ML for binding affinity predictions," *Computer Aided Drug Design Gordon Research Symposium*, 2025.
2. **Kaminow, B.**, "Creating pan-human and population-specific consensus representations of the reference genome and assessing their effect on functional genomic data analysis," *Genome Informatics*, 2019.